

Exercise 1: Charging current of an underground cable

Consider the parameters of the 150 kV underground cable that are specified in Table 3.2.

- Compute the maximum current that is allowed to flow in this cable.
- Compute the current required to charge the distributed capacitance of the cable
- Compute the length, at which the charging current at the sending end of the cable matches its maximum current.

Table 3.2 Comparison of typical parameters of a 150 kV overhead line and underground cable.

Power carrier characteristics	Overhead line	Underground cable
Conductor material	copper	copper
Conductor area [mm ²]	150	240
Rated frequency [Hz]	50	50
Rated voltage [kV]	150	150
Rated power [MVA]	130	135
r [Ω /km]	0.125	0.12
x [Ω /km]	0.425	0.166
c [nF/km]	11.7	210

Consider the parameters of the 150 kV underground cable that are specified in Table 3.2. From the rated power, we can compute the maximum current that is allowed to flow in this cable:

$$|I_{\max}| = \frac{135 \times 10^6}{\sqrt{3} \cdot 150 \times 10^3} = 520 \text{ A} \quad (3.18)$$

The current required to charge the distributed capacitance of the cable, amounts:

$$|I_c| = \omega c |V_{LN}| = 2\pi \cdot 50 \cdot 210 \times 10^{-9} \cdot \frac{150 \times 10^3}{\sqrt{3}} = 5.7 \text{ A/km} \quad (3.19)$$

If this cable is *unloaded* and $520/5.7 = 91$ km long, the maximum allowed current has been reached at the sending end of the cable!

(Van der Sluis 2008)